Major Components

Ford Ecat Electronic Parts Catalog

Major Components

This transmission has the following major components:

- Transmission case with integral torque converter housing
- Torque converter
- Fluid pump
- Three drive clutches:
  - Forward clutch (A)
  - Direct clutch (B)
  - Overdrive clutch (E)
- Three brake clutches:
  - Intermediate clutch (C)
  - Low/reverse clutch (D)
  - One-Way Clutch (OWC)
- Two planetary gearsets:
  - Front single planetary gearset (1 sun gear, 1 carrier, 1 ring gear)
  - Rear ravigneaux planetary gearset (2 sun gears, 1 carrier with 2 sets of pinion gears and 1 ring gear)
- Main control:
  - Upper valve body
  - Lower valve body
  - Transmission Range (TR) sensor
  - Turbine Shaft Speed (TSS) sensor
  - Output Shaft Speed (OSS) sensor
  - Transmission Fluid Temperature (TFT) sensor
  - Line Pressure Control (LPC) solenoid
  - Torque Converter Clutch (TCC) solenoid
  - Five electronically controlled shift solenoids

In addition to the torque converter, the other shift elements are:

- three rotating multi-plate clutches: forward (A), direct (B) and overdrive (E).
- two fixed multi-disc brakes: intermediate (C) and low/reverse (D).

All gear shifts from 1st to 6th or from 6th to 1st are power-on overlapping shifts. That is, during the shift, one of the clutches must continue to transmit the drive at lower main pressure until the other clutch is able to accept the input torque.

The shift elements, clutches or brakes are engaged hydraulically. The transmission fluid pressure builds up between the cylinder and the piston, pressing the clutches together.

The purpose of these shift elements is to carry out in-load shifts with no interruption to traction.

Multi-plate clutches; forward, direct and overdrive, supply power from the engine to the planetary geartrain. Multi-disc brakes, intermediate and low/reverse, press against the transmission housing in order to achieve a torque reaction effect.

Multi-Plate Clutch

Clutches; overdrive (E), forward (A) and direct (B) are balanced in terms of dynamic pressure. That is, their pistons are exposed to the transmission fluid flow on both sides, in order to prevent pressure buildup in the clutch as speed increases.

This equalization process is achieved by a baffle plate and pressure-free transmission fluid supplied by a lubricating passage, through which the space between piston and baffle plate is filled with transmission fluid.

The advantages of this dynamic pressure equalization are:

- reliable clutch engagement and release in all speed ranges.
- improved shift refinement.

Shift Overlap Control

The electro-hydraulic shift action is obtained by means of various valves in the main control valve body, actuated by pressure regulators. They engage or disengage the relevant clutches or brakes at the correct moments.

Hydraulic Systems

Fluid Pump

The torque converter is supported in the fluid pump by a needle roller bearing. The fluid pump is driven directly from the engine by the torque converter shell and supplies transmission fluid to the transmission and the hydraulic control unit.

The fluid pump draws in transmission fluid through a transmission fluid filter and delivers it at high pressure to the main pressure valve in the main control valve body unit. The valve adjusts the pressure and returns excess transmission fluid to the transmission fluid pan.

Fluid Filter

All transmission fluid that is picked up from the transmission fluid pan passes through the transmission fluid filter. The transmission fluid filter and its accompanying seal are part of the transmission fluid path from the pan to the transmission fluid pump.

Single Planetary Gearset

The single planetary gear overdrive carrier is driven by the input shaft. The single planetary gear set consists of the following components:

- One sun gear
- Four planetary gears meshing with the sun gear
- One planetary carrier
- One ring gear

Ravigneaux Planetary Gearset

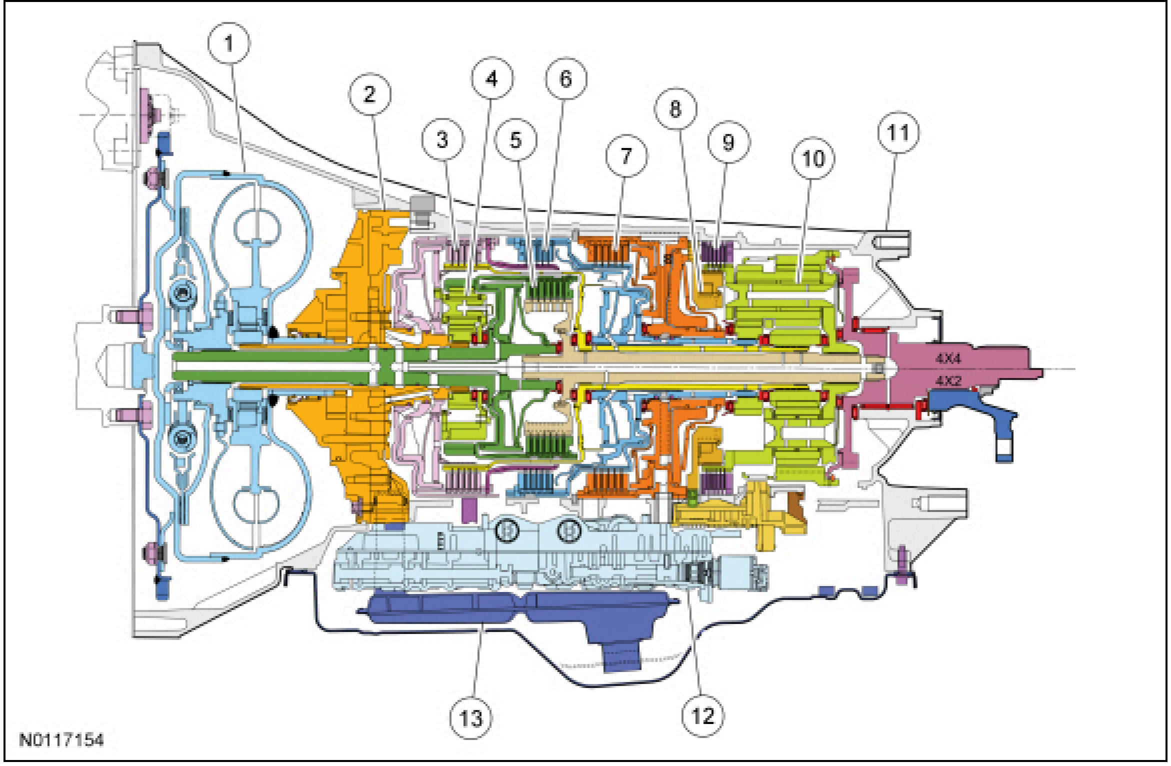
The ravigneaux planetary gearset is splined to the output shaft and consists of the following components:

- Two sun gears of different sizes
- Three long planetary gear pinions meshing with the large sun gear and the ring gear
- Three short planetary gear pinions meshing with the small sun gear and the long pinions
- One planetary carrier
- One ring gear

Output Shaft

The output shaft provides torque to the driveshaft and rear axle assembly. It is splined to the ring gear of the rear/ravigneaux planetary gearset.

Major Components Cutaway View



Item	Description
1	Torque converter
2	Front pump assembly
3	Forward clutch (A)
4	Front planetary gearset
5	Overdrive clutch (E)
6	Direct clutch (B)
7	Intermediate clutch (C)
8	One-Way Clutch (OWC)
9	Low/reverse clutch (D)
10	Rear planetary gearset
11	Transmission case
12	Main control assembly
13	Transmission fluid filter